Evaluating the Mashgin Checkout System:

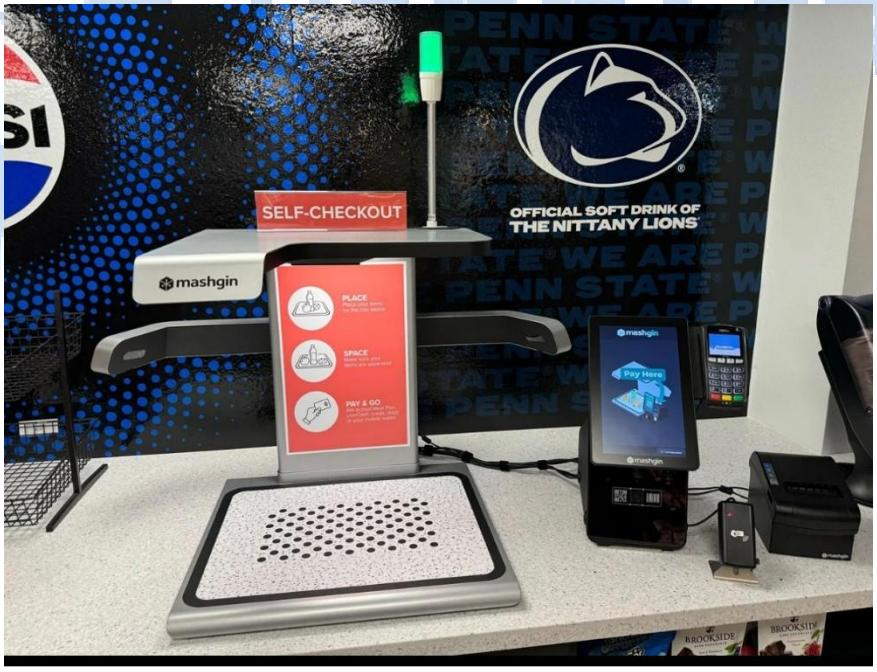
A Performance Analysis of ABS Convenience Store Operations

BAN 888: Implementing Analytics for Business

Penn State Smeal College of Business

Part I - Business Problem Framing and Analytics Problem Framing

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1. EXECUTIVE SUMMARY

This project asks a practical question: did the Mashgin checkout system create measurable operating value for Penn State Auxiliary and Business Services convenience stores, or are the observed changes mostly explained by outside factors such as price increases and nearby population changes? The answer matters because Mashgin is more than a technology purchase. It changes checkout speed, staffing patterns, customer experience, and the way store managers can influence sales through placement, bundling, and promotions.

The preliminary analysis compares April 2023 legacy checkout performance to April 2026 Mashgin performance for four comparable convenience store areas: South, East, Pollock, and West. For deeper analysis, we have a weekly breakdown of transactional data for those locations starting in August 2022. A fifth location, North, is excluded for now because its convenience store sales cannot yet be separated cleanly from full-service dining activity. A hypothesis test suggests that the stores appear to be handling more purchases, but customers are spending less per purchase.

This finding points to two decisions for ABS. First, leadership needs to know whether the transaction increase is real after controlling for price and population. Second, managers need to know how to capture more value from each faster checkout visit. The project will use an ARIMAX weekly time-series model and Apriori market basket analysis. The hypothesis test gives an initial performance verdict, but ARIMAX will illustrate whether the effect persists across weeks as the new systems come online. And Apriori identifies item combinations that can support product placement, bundling, stocking, and promotion decisions.

2. BUSINESS PROBLEM AND ORGANIZATIONAL VALUE

In 2024, ABS began to phase in Mashgin, an AI-powered scanning payment system, at its University Park convenience store network. Customers place all items on the scanning surface, the system identifies the items, and the customer completes payment. Instead of scanning every item manually, a Customer Service Representative can monitor multiple units, help with exceptions, and support payment compliance. The expected value is faster checkout, higher throughput during peak periods, and better use of staff time.

Stakeholders include ABS leadership, dining and convenience store managers, store attendants, and the campus customers who use the stores. Leadership needs to know whether the investment is producing a return. Managers need to know whether the system changed transaction volume, sales, and shopping behavior. Store attendants are affected because their role shifts from register operation to customer support and monitoring. Students and campus customers are affected because checkout speed, product availability, and store layout shape the convenience store experience.

The project creates value by connecting the system change to real operating decisions. If Mashgin increased transaction capacity, ABS can use that evidence to support continued deployment and targeted improvements. If transactions have increased but average transaction sale amount has gone down, perhaps managers can respond with layout changes, product adjacency, bundles, or promotions. Apriori matters in this section because it provides the practical bridge from performance measurement to store action: it can show which products naturally sell together and where the store has an opportunity to encourage a larger transaction without slowing checkout.

The goal is not just to prove whether Mashgin worked. It is to give leadership practical recommendations and incentives. The performance models should answer whether the system is

associated with improved sales. The market basket analysis answers what ABS can do with that information at the item placement level.

3. ANALYTICS PROBLEM FRAMING AND PRELIMINARY ANALYSIS

The analytics problem has two linked parts. The first is a performance evaluation: did the new checkout system change store performance after adjusting for price and population? The second is a customer basket question: if checkout is faster and customers are making more frequent but smaller purchases, which product combinations can help ABS improve average transaction value? These are connected because a faster checkout system creates value only if ABS can preserve or improve the value of each visit while also increasing volume.

The first statistical test shows performance before and after Mashgin deployment. The initial question of change focuses on April 2023 and April 2026 for South, East, Pollock, and West. Performance measures include transaction count, total sales, price-adjusted sales, population-adjusted transactions, and average transaction value. Then we can focus on noticeable transitions with the ARIMAX model, and look at forecasting that is adjusted for factors such as price and population changes.

Apriori is important because it addresses the question that the hypothesis test and ARIMAX model cannot answer by themselves. A performance model can show that average transaction value went down or that transaction count went up, but it cannot tell a store manager which items should be placed near each other or which combinations are strong candidates for bundles. Apriori uses transaction-level item combinations to identify common baskets, product affinities, and high-lift relationships that can be converted into store-level tests.

In sum, the hypothesis test will evaluate whether adjusted post-Mashgin lift differs meaningfully from operations prior ot Mashgin. The ARIMAX model strengthens the analysis by

using weekly observations over time instead of relying only on one before-and-after comparison. And the Apriori will give insights into item buying patterns.

4. DATA SOURCES, CLEANING, AND PROJECT FEASIBILITY

The first phase draws on three main sources. Legacy transactions from the MICROS system are stored in a SQL database and can be queried directly for historical sales, transaction counts, dates, and locations. Mashgin data is available from vendor systems, but the team is still working through a reliable feed. And nearby residential population data from contract data will be linked to the store areas most likely to serve those customers so that changes in the surrounding customer base are not mistaken for system performance.

The Mashgin data requires special cleaning attention. API calls are being developed, but there have been cases where the vendor feed stopped returning complete data while still producing success messages. That creates a data risk because a technically successful response may still be incomplete. ABS has asked the vendor for a flat file feed so the data can be reconciled, reloaded, and audited more easily, and that is in progress.

As an interim solution, the team inspected the Mashgin web reporting application and found that the site assembles sales aggregates using JSON response data. The JSON responses were saved into text files, and a custom C# application was built to read those files and load the results into SQL. This allows the team to validate, aggregate, and retrieve Mashgin reporting data while the formal feed is being resolved. Weekly totals from this interim SQL load will be reconciled against the Mashgin reporting interface before analysis.

The cleaned performance dataset will be a weekly store-level panel. Each row will represent one store and one week per system. WeekStartDate will be the anchor field, and Year, Semester, and WeekNumber will be calculated from that date rather than typed manually. That

creates a simple quality check: if a row is labeled 2023 but the week date falls in 2026, the row can be flagged before modeling. This dataset supports the hypothesis test and ARIMAX model.

Table 1. Example of weekly dataset

Field	Purpose	Calculation or source
StoreID / StoreName	Location key.	Standard crosswalk between legacy POS and Mashgin names.
WeekStartDate / WeekEndDate	Weekly observation.	Consistent week start; derives Year, Semester, WeekNumber.
TotalSales	Nominal weekly sales.	Sum of completed transaction dollars.
TransactionCount	Checkout volume.	Count of completed transactions or checks.
AvgTransactionValue	Average spend per transaction.	TotalSales / TransactionCount.
LegacyPOSIndicator	Pre-Mashgin operation.	1 for legacy POS weeks; otherwise 0.
MashginIndicator	Mashgin operation.	1 for Mashgin weeks; otherwise 0.
PriceTo2026Multiplier	Controls for price changes.	Base semester = 1.000; later semesters based on matched item prices.
PriceAdjustedSales	Comparable-dollar sales.	TotalSales * PriceTo2026Multiplier.
PopulationCount	Nearby customer base.	Residence hall or assigned population for the store area.
PopulationTo2026Multiplier	Controls for population change.	TargetPopulation2026 / CurrentPopulation
PopulationAdjustedTransactions	Normalizes transactions.	TransactionCount * PopulationTo2026Multiplierr.
FullyAdjustedSales	Controls for price and population.	TotalSales * PriceTo2026Multiplier * PopulationTo2026Multiplier.

The price index will be calculated at the semester level because ABS prices are generally set by semester. The base semester, such as Spring 2023, will equal 1.000. Later semesters will be compared against the base using matched top-selling items. The preferred method is a weighted comparable-item index, where item weights come from the base-period sales mix. If complete

weights are unavailable, the team can use a matched top-item average or median price-change index and document that limitation. Price changes occur from semester to semester, so a semester price index could be introduced as a multiplier so we can compare August 2022 to April 2026, for example.

The population multiplier will be calculated by linking each store to its surrounding residence hall or nearby customer population. The target comparison period will be set to 1.000. For each earlier period, the multiplier will equal the target-period nearby population divided by that period's nearby population. For example, if a store area had 2,000 nearby residents in an earlier period and 2,100 nearby residents in the target period, the population multiplier would be 1.050. Earlier transaction counts can then be multiplied by 1.050 to express activity on a target-period population basis, or the population measure can be included directly as an explanatory variable in the ARIMAX model. Either approach helps prevent customer-base growth from being mistaken for a Mashgin effect.

The Apriori input will be cleaned differently from the weekly performance panel. It will use a transaction-item structure from the Mashgin item-level data, with fields such as TransactionID, StoreID, BusinessDate, ItemID or ItemName, quantity, and line amount. The cleaning step will remove voids, refunds, test transactions, non-merchandise lines, and duplicate item records. The data will then be converted into one basket per transaction, where each basket contains the set of items purchased together. This allows the team to calculate frequent itemsets and association rules even if full historical basket-count fields are not available.

5. PROPOSED ANALYTICAL APPROACH

The first check is a matched pairs hypothesis test using adjusted location-level lift values. The analysis compares observed post-Mashgin performance against an expected baseline that

accounts for price and population changes. This follows the logic of index-based same-store comparison: outside factors are controlled before performance differences are attributed to the checkout system. The statistical test uses the one-sample t-test for determining whether a sample mean differs from a hypothesized value of zero, applied to location-level adjusted lift values.

The second method is ARIMAX, which extends the analysis to a weekly time-series model. ARIMAX is useful because store performance is related to prior-week patterns and outside conditions. The model will use weekly store-level observations with external variables such as MashginIndicator, PriceMultiplier, PopulationMultiplier, location, semester, and possibly academic calendar flags that include anomalies like Spring Break time.

The third method is Apriori association rule mining using April 2026 Mashgin item-level transaction data. Apriori identifies item sets that occur together above a minimum support threshold and generates rules measured by support, confidence, and lift. Support shows how common an item combination is. Confidence shows how often one item predicts another. Lift shows whether the relationship is stronger than chance. Lift greater than one indicates that the items occur together more often than expected if purchases were independent.

6. EXPECTED RECOMMENDATIONS, DELIVERABLES, AND NEXT STEPS

This project expects to deliver some practical information to ABS. First, it will provide a performance verdict on the Mashgin transition using adjusted hypothesis testing and weekly ARIMAX modeling. Second, it will create a clean weekly monitoring dataset that ABS can reuse after the class project. Third, it will provide product placement and bundling guidance based on Apriori association rules. And finally, we can research and recommend some approaches, such as incentives for managers when it comes to store performance.

The next steps are to finalize the weekly data, complete the price and population indexes, run the adjusted hypothesis test, and then fit the ARIMAX model. In parallel, the team can prepare the Apriori input file by converting Mashgin item-level transaction data into baskets and screening itemsets for enough support to be operationally meaningful. Together, these steps will give ABS a practical answer to the core business question: whether Mashgin improved performance, where the improvement is strongest, and what actions can increase the value of each customer visit.

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